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ARTIFICIAL INTELLIGENCE & ML

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Designed for University Exams & AI/ML Placements

CHAPTER 1: INTRODUCTION TO AI & MACHINE LEARNING



What is Artificial Intelligence?

AI is the simulation of human intelligence in machines programmed to think, reason, and learn from experience.

```
# Simple Linear Regression with Scikit-Learn
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1], [2], [3], [4]])
y = np.array([2, 4, 6, 8])

model = LinearRegression()
model.fit(X, y)
print("Prediction for 5:", model.predict([[5]])) # Output: [10.]
```